

Economics 655 – Fall 2013
Experimental Economics
TH 10:20 am-11:35 am
Room 3095, Allen Building

Dr. Alex Brown
Office Hours: TH 11:30am-1:00 pm
Room 3056, Allen Building
abrown@econmail.tamu.edu
Blackboard page at: elearning.tamu.edu

Purpose and Expectations:

This course will prepare students to be able to design, implement and analyze their own economics experiments. In addition, the course will familiarize students with previous work in selected topics in experimental economics. The final assignment of this class will be a written assignment where each student will design an experiment. Successful students will leave this course with a greater familiarity with experimental methods and previous research in economics, as well as a experimental design that when implemented could be turned into a significant contribution to the discipline. Students that produce outstanding experimental designs will receive funding for their experiment after the class has completed.

Prerequisites:

This is a graduate level class for economics students that have completed their first year. I expect students to have knowledge of economic theory and econometrics consistent with this level.

Texts:

I have notified the university bookstores of each required and recommended text. If you are unable to get a copy of any of the books, I suggest you order online today. The library also has each book available on course reserves. The first parts of each book are available from the library as a .pdf (<http://library-reserves.tamu.edu/areslocal/index.htm>), so that you may study as you wait to have a book delivered. The .pdfs do not span the entire book so you will need to buy them as we proceed further with the class.

Required:

John H. Kagel and Alvin Roth, *Handbook of Experimental Economics*, Princeton University Press, ISBN 978-0691058979.

Daniel Friedman and Shyam Sunder, *Experimental Methods: A Primer for Economists*, Cambridge University Press, ISBN 978-0521456821.

Recommended:

Douglas D. Davis and Charles A. Holt, *Experimental Economics*. Princeton University Press, ISBN 978-0-691043173.

Course Policies:

Grading:

- Students will earn numerical grades using the following weighting:

In Class Quizzes/Responses/Discussion	20%
Written Responses	20%
Class Lecture and Written Summary	20%
Final paper	30%
Final presentation	10%

- Students will earn letter grades using these standards:

A:

- Final numerical grade above 90%
- All four (class lecture, written summary, final paper, final presentation) writing assignments completed **ON TIME** (or lateness excused), all feedback (marked up rough drafts, proposals) picked up by student.

B:

- Final numerical grade above 80%
- All four (class lecture, written summary, final paper, final presentation) writing assignments completed **ON TIME** (or lateness excused), all feedback (marked up rough drafts, proposals) picked up by student.

C:

- Final numerical grade above 70%
- All four (class lecture, written summary, final paper, final presentation) writing assignments completed **ON TIME** (or lateness excused), all feedback (marked up rough drafts, proposals) picked up by student.

D:

- Final numerical grade above 60%
- All four (class lecture, written summary, final paper, final presentation) writing assignments completed **ON TIME** (or lateness excused), all feedback (marked up rough drafts, proposals) picked up by student.

This scale does not apply to students who violate the honor code in their work for this class. The standard grade for such violations is F*. I will report all violations to the Honor Council.

Notes:

- As the above scale indicates, failure to complete any part of the writing process (class lecture, written summary, final paper, final presentation) **ON TIME** will result in failure for the class.
- There is no extra credit or extra work to improve your grade. Nothing will be curved.

In Class Quizzes/Responses/Discussion:

- Each week of class, usually on Tuesday (i.e., **8/29**, 9/03, 9/10, 9/17, 9/24, 10/01, 10/08, 10/15, 10/22, **10/31**, **11/07**, **11/14**) we will begin class with an informal quiz and discussion of research ideas.
- Quiz may be written or oral. Written quizzes will be evaluated separately.
- Unless it is accompanied by a university-approved and documented excuse, missing class will count as a 0.

Written Responses:

- Each week of class, usually on Tuesday (i.e., **8/29**, 9/03, 9/10, 9/17, 9/24, 10/01, 10/08, 10/15, 10/22, **10/31**, **11/07**, **11/14**) you will turn in 2, 1-page (250 words) typed responses.
- The first will be a summary and analysis of the readings for that week.
- The second will be an experimental research idea. You may represent the same idea each week provided you have sufficiently developed it from your last week's response.
- Unless it is accompanied by a university-approved and documented excuse, late homework officially counts as a 0.

- You may *discuss* homework with each other, but you *must* write up each response *individually*. Copying work is prohibited and violates the Honor Code.

Class Lecture and Written Summary:

- Each student will be expected to take part leading the *second* class in one of nine lecture topics (9/19, 9/26, 10/3, 10/10, 10/17, **10/29, 11/5, 11/12, 11/19**)
- Class lectures will be assigned during the second week of class.
- Students will choose some number of academic papers related to the topic and provide a 5-page summary (1250 words) of those works.
- That summary will be due *in the preceding class*.
- The summary will be distributed to all students to read before the class lecture.
- During the class lecture the student will present those papers.
- If we have more student than classes, some classes will involve multiple students. However each student must submit his/her own unique 5 page summary.

Final paper:

- Students will complete a 10-page (2500 word) final paper due on the last day of class, December 3.
- The paper should resemble academic work on experimental economics, with the exception that the paper will be incomplete. Students will not be required to have run the experiment designed in their paper.
- Thus, the introduction, related literature, hypothesis/theory, and procedure sections should be completed. The results and conclusion sections should be hypothetical based upon the possible results of the experiment.
- Students may collaborate on their final paper. All collaborations require my approval.

Final presentation:

- Beginning November 21, students will present their experiment designs to the class in 25-minute presentations.
- Presentation will be scheduled during the second week of class.
- Students need not turn in their paper before their presentation.
- In the event of collaboration, I will choose the person who will present the research on the day of the presentation. My intent is to pick the collaborator who looks least prepared.

Attendance and Conduct:

- I expect you to attend class, and to have done the readings before class.
- I expect you to be respectful to your fellow students and me. This means
 - You will show up to class on time.
 - You will attend all classes, especially those led by your fellow students.

Office Hours:

- My official office hours are 11:30am-1:00pm on Tuesday and Thursday.
- Please do not show up to my office at other times without an appointment.
- You may email me to schedule appointments for office hours at other times.

Rule 7-Absences:

- I designed my syllabus and course policies to be consistent with Texas A&M University's student rule 7 on student absences.
- If any part of this syllabus contradicts rule 7, that part will be voided and rule 7 will take

precedent.

- For any part of rule 7 that my syllabus does not specifically address, you may assume that rule 7 will be applied in that situation.

Academic Integrity Statements:

AGGIE HONOR CODE

“An Aggie does not lie, cheat, or steal or tolerate those who do.”

Upon accepting admission to Texas A&M University, a student immediately assumes a commitment to uphold the Honor Code, to accept responsibility for learning, and to follow the philosophy and rules of the Honor System. Students will be required to state their commitment on examinations, research papers, and other academic work. Ignorance of the rules does not exclude any member of the TAMU community from the requirements or the processes of the Honor System.

For additional information please visit: www.tamu.edu/aggiehonor/

Americans with Disability Act (ADA) Policy Statement:

The Americans with Disabilities Act (ADA) is a federal anti-discrimination statute that provides comprehensive civil rights protection for persons with disabilities. Among other things, this legislation requires that all students with disabilities be guaranteed a learning environment that provides for reasonable accommodation of their disabilities. If you believe you have a disability requiring accommodation, please advise the instructor and contact the Department of Student Life/Services for Students with Disabilities, in Cain Hall or call 845-1637.

Course Schedule:

This schedule is my best guess on how the class lectures will unfold. I will make every effort to follow this guide. However, there are several circumstances beyond my control that can affect the timing of class (e.g., hurricanes, snow, fire drills). While the following is a good guide for the class, it is *extremely unlikely* that everything will go as planned. Therefore **DO NOT** trust this guide solely for class content, deadlines, and test days. This guide **DOES NOT** exclude each student from their responsibility to know what transpires and what will transpire on each day of class.

Class	Date	Topic	Homework Due
1	8/27	Syllabus/Introduction	-
2	8/29	History of Experimental Economics	Readings, 1-page response & 1-page design
3	9/03	Methodology of Experimental Economics	Readings, 1-page response & 1-page design
4	9/05		-
5	9/10	Software in Experimental Economics	Readings, 1-page response & 1-page design
6	9/12	Ethics and the IRB	CITI training course completed
7	9/17	Asset Markets	Readings, 1-page response & 1-page design
8	9/19		*

Class	Date	Topic	Homework Due
9	9/24	Auctions	Readings, 1-page response & 1-page design
10	9/26		*
11	10/01	Public Goods	Readings, 1-page response & 1-page design
12	10/03		*
13	10/08	Game Theory	Readings, 1-page response & 1-page design
14	10/10		*
15	10/15	Individual Decision Making/Preferences	Readings, 1-page response & 1-page design
16	10/17		*
17	10/22	Elicitation Mechanisms	Readings, 1-page response & 1-page design
	10/24	NO CLASS: North American Economic Science Association Meetings	
18	10/29	Elicitation Mechanisms	*
19	10/31	Behavioral Finance	Readings, 1-page response & 1-page design
20	11/05		*
21	11/07	Other Regarding Preferences	Readings, 1-page response & 1-page design
22	11/12		*
23	11/14	Special Topics: Field Experiments/Neuroeconomics	Readings, 1-page response & 1-page design
24	11/19		*
25	11/21	Student Presentations	-
26	11/26		-
27	12/03		FINAL PAPERS DUE
28	12/06	Student Presentations (12:30-2:30 pm, if necessary)	-

*A selected student(s) will lead lectures on these days. Additionally each student leading the lecture will submit a 5-page summary of the readings in the previous class.

Selected Readings:

Week 1a: Introduction

Friedman and Sunder, Ch. 1 and 2.4

Kagel and Roth, 1.II (21-23)

Plott, Charles R. (1982) "Industrial Organization Theory and Experimental Economics," *Journal of Economic Literature* 20(4): 1485-1527. (read pages 1519-1523)

Optional: Roth, Alvin E. *Laboratory Experimentation in Economics: Six points of view*. Cambridge University Press, New York. (read Ch. 1 & 7)

Week 1b: History of Experimental Economics

Friedman and Sunder, 2.6

Kagel and Roth 1-19, 49-60, 361-365, 368-372

Chamberlin, E. H. (1948) "An Experimental Imperfect Market," *Journal of Political Economy* 56(2): 95-108.

Smith V. (1962) "An Experimental Study of Competitive Market Behavior," *Journal of Political Economy* 70(2): 111-137.

Plott C. and V. Smith (1978) "An Experimental Examination of Two Exchange Institutions," *Review of Economic Studies*, 45(1): 133-153.

Week 2: Experimental Methods I

Friedman and Sunder, Ch 2-7.

Davis and Holt 1.6 (19-33).

Davis and Holt Ch. 9

Optional: Smith (1982) "Microeconomic Systems as an Experimental Science." *American Economic Review*, 72(5): 923-955.

Week 3: Experimental Methods II

Fischbacher, Urs (2007). "z-Tree: Zurich Toolbox for Ready-Made Economic Experiments." *Experimental Economics*. 10(2): 171-178.

Fischbacher, Urs (2006) *Z-tree 2.1 Reference Manual*. Available at:
<http://www.iew.uzh.ch/ztree/ztree21ref.pdf>

Fischbacher, Urs (2002) *Z-tree Tutorial*. Available at: <http://www.iew.uzh.ch/ztree/ztree21tutorial.pdf>

U.S. Department of Health and Human Services. The Belmont Report. Available at:
<http://www.hhs.gov/ohrp/humansubjects/guidance/belmont.html>

Complete CITI training for certification at TAMU: <http://rcb.tamu.edu/humansubjects/training>

Week 4: Asset Markets

Kagel and Roth, Chapter 6.

Plott, Charles R., and Shyam Sunder (1982) "Efficiency of Experimental Security Markets with Insider Information: An Application of Rational-Expectations Models," *Journal of Political Economy*, 90(4): 663-698

Plott, Charles R., and Shyam Sunder (1988) "Rational Expectations and the Aggregation of Diverse Information in Laboratory Security Markets," *Econometrica*, 56(5): 1085-1118.

Smith, Vernon L., Gerry L. Suchanek, and Arlington W. Williams (1988) "Bubbles, Crashes, and Endogenous Expectations in Experimental Spot Asset Markets," *Econometrica*, 56(5): 1119-1151.

Bossaerts, P., and C. Plott, (2004), "Basic Principles of Asset Pricing Theory: Evidence from Large- Scale Experimental Financial Markets," *Review of Finance*, 8(2): 135-169.

Week 5: Auctions

Kagel and Roth, Chapter 7.

Kagel and Roth II, Chapter 10. Available at
http://www.econ.ohio-state.edu/kagel/Auctions_Handbook_vol2.pdf

Kagel, J. and Levin, D. (1986) "The Winner's Curse and Public Information in Common Value Auctions," *American Economic Review*, 76(5): 894-920.

Kagel, J. and Levin, D. (1999) "Common Value Auctions with Insider Information," *Econometrica*, 67(5): 1219-1238.

Brown, Plott and Sullivan (2009) "Collusion Facilitating and Collusion Breaking Power of Simultaneous Ascending Price and Descending Price Auctions." *Economic Inquiry*, 47(3): 395-424.

Week 7: Public Goods

Kagel and Roth, Chapter 2.

- Andreoni, J. (1995). "Warm Glow vs. Cold Prickle: The Effect of Positive and Negative Framing on Cooperation in Experiments." *Quarterly Journal of Economics* 110(1): 1-21.
- Fehr, E. and S. Gächter (2000). "Cooperation and Punishment in Public Goods Experiments." *American Economic Review* 90(4): 980-994.
- Ostrom, Elinor and Harini Nagendra (2006). "Insights on Linking Forests, Trees, and People from the Air, on the Ground, and in the Laboratory." *Proceedings of the National Academy of Sciences* 103(51): 19224-31.
- Ashley, R., S. Ball and C. Eckel (2010). "Motives for Giving: A Reanalysis of Two Classic Public Goods Experiments." *Southern Economic Journal* 77(1): 15-26.

Week 6: Game Theory

Davis and Holt Ch. 2, 90-110.

- Forsythe, R., R. M. Isaac and T. R. Palfrey (1989). "Theories and Tests of "Blind Bidding" in Sealed-bid Auctions." *The RAND Journal of Economics* 20(2): 214-238.
- Van Huyck, J., Battalio, R., and Beil, R. (1990) "Tacit Coordination Games, Strategic Uncertainty, and Coordination Failure," *American Economic Review*, 80(1): 234-248.
- Ho, T, Camerer, C. and Weigelt, K. (1998) "Iterated Dominance and Iterated Best Response in Experimental P-Beauty Contests," *American Economic Review*, 88(4): 947-969.
- Palacios-Huerta, I. and O. Volij (2009). "Field Centipedes." *American Economic Review* 99(4): 1619-1635.

Week 8: Individual Decision Making/Preferences

Kagel and Roth, Chapter 8.

- Kahneman, D. and A. Tversky (1979). "Prospect Theory: An Analysis of Decision under Risk." *Econometrica* 47(2): 263-291.
- Kahneman, D., J. L. Knetsch and R. H. Thaler (1990). "Experimental Tests of the Endowment Effect and the Coase Theorem." *Journal of Political Economy* 98(6): 1325-1348.
- Andersen, S., G Harrison, M Lau, E. Rutström, (2008). "Eliciting Risk and Time Preferences." *Econometrica*, 76, 583-618.
- Brown, A. L., Z. E. Chua and C. F. Camerer (2009). "Learning and Visceral Temptation in Dynamic Saving Experiments." *Quarterly Journal of Economics* 124(1): 197-231.

Week 9: Eliciting Individual Preferences

- Hoffman, E., K. McCabe, K. Shachat and V. Smith (1994). "Preferences, Property Rights, and Anonymity in Bargaining Games." *Games and Economic Behavior* 7(3): 346-380.
- Optional: Hoffman, E., K. McCabe and L. S. Vernon (1996). "Social Distance and Other-Regarding Behavior in Dictator Games." *American Economic Review* 86(3): 653-660.
- List, J.A. (2007). "On The Interpretation Of Giving In Dictator Games." *Journal of Political Economy* 115: 482-492.
- Holt, C. A. and S. K. Laury (2002). "Risk Aversion and Incentive Effects." *American Economic Review* 92(5): 1644-1655.
- Dave, C., C. Eckel, C. Johnson and C. Rojas (2010). "Eliciting risk preferences: When is simple better?" *Journal of Risk and Uncertainty* 41(3): 219-243.
- Andreoni, J and C. Sprenger. "Estimating Time Preferences from Convex Budgets." *American Economic Review* 102(7): 3333-56.

Week 10: Behavioral Finance

- Bloomfield, R. and J. Hales (2002). "Predicting the next step of a random walk: experimental evidence of regime-shifting beliefs." *Journal of Financial Economics* 65(3): 397-414.
- Kirchler, E., B. Maciejovsky and M. Weber (2005). "Framing Effects, Selective Information, and Market Behavior: An Experimental Analysis." *Journal of Behavioral Finance* 6(2): 90-100.
- Fellner, G. (2009). "Illusion of Control as a Source of Poor Diversification: Experimental Evidence." *Journal of Behavioral Finance* 10(1): 55-67.
- Biais, B. and M. Weber (2009). "Hindsight Bias, Risk Perception, and Investment Performance." *Management Science* 55(6): 1018-1029.
- Bossaerts, P., P. Ghirardato, S. Guarnaschelli and W. R. Zame (2010). "Ambiguity in Asset Markets: Theory and Experiment." *Review of Financial Studies* 23(4): 1325-1359.

Week 11: Other Regarding Preferences

- Fehr, Ernst & Schmidt, Klaus (1999), "A Theory of Fairness, Competition, and Cooperation." *Quarterly Journal of Economics*, 114, 817-868.
- Andreoni, J. and Miller, J. (2002), "Giving according to GARP," An Experimental Test of the Consistency of Preferences for Altruism," *Econometrica*, 70, 737-753.
- Charness, Gary & Matthew Rabin (2002), "Understanding Social Preferences with Simple Tests," *Quarterly Journal of Economics*, 117, 817-869.
- Engelmann, D. and Strobel, M. (2004), "Inequality Aversion, Efficiency and Maximim Preferences in Simple Distribution Experiments," *American Economic Review*, 94, 857-869.
- Fisman, R., Kariv, S., & Markovits, D. (2007), "Individual Preferences for Giving," *American Economic Review*, 97, 1858-1876.

Week 12: Special Topics: Field Experiments/Neuroeconomics

- Kagel and Roth II, Chapter 8, available at:
<http://kuznets.fas.harvard.edu/~aroth/papers/Neuroeconomics%20rough%20draft%202011.pdf>
- Levitt, S. D. and J. A. List (2009). "Field experiments in economics: The past, the present, and the future." *European Economic Review* 53(1): 1-18.
- Bechara, A., H. Damasio, D. Tranel and A. R. Damasio (1997). "Deciding Advantageously Before Knowing the Advantageous Strategy." *Science* 275(5304): 1293-1295.
- Plassmann, H., J. O'Doherty, B. Shiv and A. Rangel (2008). "Marketing actions can modulate neural representations of experienced pleasantness." *Proceedings of the National Academy of Sciences* 105(3): 1050-1054.
- Charness, G. and U. Gneezy (2009). "Incentives to Exercise." *Econometrica* 77(3): 909-931.